

Final exam, July 2026

KSE, Algebraic topology, Summer 2026

Length: 3 hours

Credits: 40 points

Notation and prerequisites:

- For all $n \geq 1$, we choose a generator $[S^n]$ of the group $H_n(S^n, \mathbb{Z}) \simeq \mathbb{Z}$. If $f : S^n \rightarrow S^n$ is a continuous map, recall that the *degree of f* , denoted $\deg(f)$, is defined as the integer such that $f_*([S^n]) = \deg(f)[S^n]$. You can use the fact that the degree of a reflection (symmetry with respect to some hyperplane of $\mathbb{R}^{n+1}$) of S^n is -1 .
- An automorphism f of the topological group $T \simeq \mathbb{R}^2/\mathbb{Z}^2$ (the torus) is always given by an element $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{Z})$ and is defined by the formula $f(x, y) = (ax + by, cx + dy)$.
- $\mathbf{F}_2$ stands for the free group generated by 2 elements.
- The integral cohomology ring of $\mathbb{C}P^2$ is given by

$$H^*(\mathbb{C}P^2, \mathbb{Z}) \simeq \mathbb{Z}[\alpha]/(\alpha^3)$$

where α lives in degree 2 and is a generator of $H^2(\mathbb{C}P^2, \mathbb{Z}) \simeq \mathbb{Z}$.

Exercise 1. Fixed point and degree of a map

Define the antipodal map $a : S^n \rightarrow S^n$ by $a(x) = -x$. Let $f : S^n \rightarrow S^n$ be a continuous map.

- (1) Assume that f has no fixed point. Show that f is homotopic to a .
- (2) What is the degree of the antipodal map a ? Deduce that if $\deg(f) \neq (-1)^{n+1}$, then f must have a fixed point.
- (3) Deduce that $\mathbb{Z}/2\mathbb{Z}$ is the only non-trivial group that can act freely on S^n when n is even (recall that an action of a group G by homeomorphisms is free if the automorphisms induced by non-trivial elements have no fixed point).

Exercise 2. About some subgroups of $\mathbf{F}_2$

Answer the following points using the theory of covering spaces.

- (1) Is any index 3 subgroup of $\mathbf{F}_2$ a normal subgroup? After answering geometrically (do not hesitate to do some drawings), give an explicit subgroup of $\mathbf{F}_2$ of index 3 which is not normal.
- (2) Is it true that for any $n > 0$ there exists a subgroup H of $\mathbf{F}_2$ which is a free group on n generators?

Exercise 3. Conditions for a continuous square root map

We cannot always find a continuous square root of a complex valued map on a topological space, and we propose here to study this obstruction. Let X be a path-connected, locally path-connected and SLSC topological space, and $f : X \rightarrow \mathbb{C}$ be a continuous map which is never 0. By a *continuous square root* of f we mean a continuous function $g : X \rightarrow \mathbb{C}$ such that $g(x)^2 = f(x)$ for any $x \in X$.

- (1) Formulate the existence of a continuous square root of f in terms of lifting of f to a covering space of $\mathbb{C}^\times$ that you should specify. Show that there exists a 2-sheeted covering $p : Y \rightarrow X$ such that $f \circ p$ has a continuous square root.
- (2) Show that f admits a continuous square root *if and only if* the covering p (from the previous question) is trivial, ie. isomorphic to $X \sqcup X \rightarrow X$.
- (3) What can you conclude about this problem when X is simply connected? When $\pi_1(X)$ has no subgroups of index 2?

Exercise 4. Automorphisms of the torus and extension to the solid torus

Let $T = S^1 \times S^1$ be the usual torus embedded in $\mathbb{R}^3$, and $P = S^1 \times D^2$ be the solid torus whose boundary is $\partial P = S^1 \times \partial D^2 = T$.

- (1) Compute the homologies $H_*(T)$ and $H_*(P)$ with coefficients in $\mathbb{Z}$.
- (2) Let $f : T \rightarrow T$ and $g : P \rightarrow P$ be two homeomorphisms. Describe $H_*(f) : H_*(T) \rightarrow H_*(T)$ and $H_*(g) : H_*(P) \rightarrow H_*(P)$. Deduce a necessary condition of homological nature (more precisely on the form of $H_1(f)$) such that a homeomorphism of T extends to a homeomorphism of P .
- (3) Let $f : T \rightarrow T$ be an automorphism (of topological groups) of T . Give a necessary and sufficient condition of homological nature such that f extends to a homeomorphism of P .

Exercise 5. Non-existence of structures of topological groups on the spheres

The goal of this exercise is to show that there are no structures of topological groups on the spheres S^{2n} for any $n > 0$.

- (1) Show that the natural map $H^*(S^n, \mathbb{Z}) \rightarrow \text{Hom}(H_*(S^n, \mathbb{Z}), \mathbb{Z})$ is an isomorphism. We write $w_n \in H^n(S^n)$ for the dual of $[S^n]$, ie. for the element defined by $[S^n] \mapsto 1$ in $H^n(S^n)$.
- (2) Show that for any $k \geq 0$, we have for homology and cohomology:

$$H_k(S^n \times S^n) \simeq \bigoplus_{i+j=k} H_i(S^n) \otimes_{\mathbb{Z}} H_j(S^n) \quad , \quad H^k(S^n \times S^n) \simeq \bigoplus_{i+j=k} H^i(S^n) \otimes_{\mathbb{Z}} H^j(S^n).$$

- (3) Show that for any continuous map $\mu : S^n \times S^n \rightarrow S^n$, there exist $\deg_1(\mu), \deg_2(\mu) \in \mathbb{Z}$ such that $\mu^*(w_n) = \deg_1(\mu)(w_n \otimes 1) + \deg_2(\mu)(1 \otimes w_n)$.
- (4) Assume that $\mu : S^n \times S^n \rightarrow S^n$ has a unity, ie. an element $e \in S^n$ such that $\mu(x, e) = \mu(e, x) = x$ for any $x \in S^n$. Show that $\deg_1(\mu) = \deg_2(\mu) = 1$.
- (5) Show that $w_n \smile w_n = 0$, and deduce from it that $\deg_1(\mu) \deg_2(\mu) = 0$ if n is even. Conclude.

Exercise 6. Hopf invariants

Let $n \geq 2$ and $f: S^{2n-1} \rightarrow S^n$ be a continuous map. Recall that $C(f) = S^n \cup_f D^{2n}$ is the mapping cone of f .

1. Using the mapping cone sequence of cohomology, show that

$$H^q(C(f)) \cong \begin{cases} \mathbb{Z}, & q = 0, n, 2n, \\ 0, & \text{otherwise.} \end{cases}$$

2. Let $\alpha \in H^n(C(f))$ be a generator and let $\beta \in H^{2n}(C(f))$ also be a generator. Show that there exists a unique integer $H(f) \in \mathbb{Z}$ such that

$$\alpha \smile \alpha = H(f)\beta.$$

This integer is called the *Hopf invariant* of f . Show that the sign of $H(f)$ depends on β but not on α , and $|H(f)|$ only depends on f itself.

3. Show that if $f \simeq g$, then $H(f) = H(g)$ for compatible choices of generators.
4. Assume n is odd. Show that $H(f) = 0$.
5. Let $\eta: S^3 \rightarrow \mathbb{C}P^1 \cong S^2$ be the Hopf map, i.e., the quotient map. Recall that $C(\eta) \cong \mathbb{C}P^2$ (it has essentially been proved in Exercise 4.37, you may use it directly); using the given description of $H^*(\mathbb{C}P^2; \mathbb{Z})$, show that $H(\eta) = 1$ for a suitable choice of generators.